

# SANTHOSH S

📍 Location: Coimbatore, Tamil Nadu

☎ +91 6379730301

✉ [santhosh.workmails@gmail.com](mailto:santhosh.workmails@gmail.com)

🌐 [www.linkedin.com/in/santhosh-electronics](https://www.linkedin.com/in/santhosh-electronics)

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## EXECUTIVE SUMMARY:

Electronics and Communication Engineering student with hands-on experience in ESP32-based IoT systems, PCB design, hardware integration, testing, and troubleshooting. Experienced in developing electronic prototypes from schematic design and PCB fabrication to assembly and testing. Interested in embedded systems, IoT, and hardware development, with a strong focus on practical learning and problem-solving.

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## TECHNICAL SKILLS:

**PCB Design:** KiCad, Schematic Design, PCB Layout, Gerber Generation

**PCB Fabrication:** FlatCAM, LinuxCNC, CNC PCB Milling

**Embedded Systems:** Arduino Uno, ESP32, IoT Fundamentals, Hardware Development

**CAD & 3D Design:** Autodesk Fusion 360, 3D Modeling, Enclosure Design

**Programming:** C, C++

**Testing & Debugging:** Hardware Testing, Troubleshooting, Prototype Development

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## PROJECTS:

### 1. Smart Irrigation System Using IoT

- Developed an IoT-based Smart Irrigation System with automatic and manual control for efficient plant watering.
- Implemented real-time monitoring of soil moisture, temperature, and humidity through an OLED display and mobile dashboard.
- Designed the complete PCB using KiCad, fabricated it using a CNC milling machine, and performed soldering, assembly, testing, and troubleshooting.
- Designed a customized 3D enclosure using Autodesk Fusion 360 and integrated the PCB and hardware to build a complete functional prototype.

### 2. AI-Powered PCB Inspection & Automated Defect Segregation — Ongoing

- Developing an AI-based PCB inspection system using YOLO for automated PCB defect detection.
  - Planning to detect missing/incorrect components, component placement, soldering defects, and PCB trace/track defects through image-based inspection.
  - Raspberry Pi is planned as the edge-processing platform for real-time PCB image inspection and defect detection.
  - Working towards automated defect segregation and real-time production monitoring based on AI inspection results.
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## EDUCATION:

**B.E. Electronics and Communication Engineering**

**2023 – 2027**

Coimbatore Institute of Engineering and Technology, Coimbatore

CGPA: 7.08 / 10